TrustMed-AI: Uncertainty-Aware Selective Prediction for Trustworthy Brain Tumor Classification from MRI

Tran Phi Hoc
FPT University, Vietnam
Email: hoctpse190186@fpt.edu.vn

Dinh Dai Loc
FPT University, Vietnam
Email: locddse190189@fpt.edu.vn

Ho Gia Phu
FPT University, Vietnam
Email: hanzopn2603@gmail.com

***Abstract*—Brain tumor classification from Magnetic Resonance Imaging (MRI) using deep learning has achieved remarkable accuracy exceeding 99%. However, these systems lack the ability to express uncertainty or abstain from prediction when encountering ambiguous cases—a critical requirement for clinical deployment. We present TrustMed-AI, an end-to-end framework integrating: (1) a dual-backbone architecture combining Swin Transformer V2 and ConvNeXt for complementary feature extraction via CLS-level attention fusion, (2) multi-task expert heads mimicking hierarchical clinical reasoning, (3) Monte Carlo Dropout uncertainty quantification with post-hoc temperature scaling calibration, and (4) a risk-aware selective prediction engine that triages cases into AUTO (autonomous diagnosis), ASSIST (specialist review), or ABSTAIN (rejected). On the Brain Tumor MRI dataset (4 classes), our system achieves competitive classification performance while providing calibrated confidence estimates and catching potential misclassifications through selective abstention. We argue that *trustworthiness*—not accuracy alone—should be the primary design objective for clinical AI systems.**

***Index Terms*—Brain tumor classification, MRI, uncertainty quantification, selective prediction, explainable AI, Monte Carlo Dropout, temperature scaling, clinical decision support**

I. Introduction

Brain tumors represent a significant global health burden, with over 120 types classified by the World Health Organization (WHO) based on cellular origin, anatomical location, and histological characteristics [1], [32]. Magnetic Resonance Imaging (MRI) serves as the non-invasive gold standard for brain tumor assessment, enabling distinction among common types including glioma, meningioma, and pituitary tumors [26], [27].

Recent deep learning advances have demonstrated extraordinary classification performance: Swin Transformer with AE-cGAN augmentation achieves 99.54% [6], attention-enhanced YOLOv7 reaches 99.50% [7], and fine-tuned VGG16 attains 99.24% [8]. However, a critical examination reveals a fundamental disconnect between laboratory metrics and clinical readiness.

A. The Accuracy-Safety Gap

Despite near-perfect accuracy, several critical issues remain unaddressed in current systems:

False positive burden. YOLOv7 exhibits a false-positive rate of 0.35 for the "no tumor" class—implying 35 out of 100 healthy individuals could receive false diagnoses [7].

Recall asymmetry. Xception networks show glioma recall of only 0.887, meaning approximately 11% of aggressive malignant tumors are missed [9].

Explainability failure. A rigorous 2025 XAI evaluation found Grad-CAM achieves only Dice = 0.33 ± 0.22 overlap with expert annotations [11], indicating AI attention regions poorly correspond to clinically relevant structures.

Domain shift. Models encode acquisition-specific features from training data, leading to performance degradation across institutions with different MRI equipment [12].

B. From Accuracy to Trustworthiness

These observations motivate our central thesis: *a clinical AI system that knows when it doesn't know is safer than one that is always confident*. Our contributions are:

- **NeuroFusionNet**: Dual-backbone (Swin-V2 + ConvNeXt) architecture with CLS-level attention fusion and multi-task expert heads mimicking hierarchical clinical reasoning.
- **Risk-Aware Selective Prediction**: Three-tier triage (AUTO/ASSIST/ABSTAIN) with clinical cost matrices penalizing high-risk misclassifications.
- **Trustworthiness Framework**: MC Dropout uncertainty, post-hoc temperature scaling [2], bootstrap confidence intervals, and McNemar's tests in a unified pipeline.
- **Clinical Demo**: Interactive system generating structured reports with morphological descriptors, expert consensus, and uncertainty communication.

II. Related Work

A. Deep Learning for Brain Tumor Classification

Deep learning has transformed medical image analysis [26], with CNN architectures such as ResNet [14], [29], DenseNet [28], and EfficientNetV2 [18] achieving strong MRI classification through fine-tuning ImageNet-pretrained features [8], [9]. Vision Transformers [15] capture global spatial relationships via self-attention [22], with Swin Transformer [16] reaching 99.54% using generative augmentation [6]. Hybrid CNN-ViT ensembles exploit complementary local-global features [13],

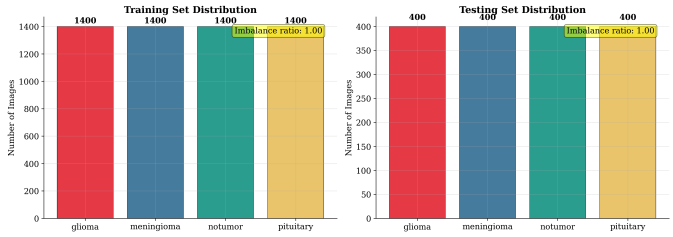


Fig. 1: Class distribution across train/validation/test splits. Stratified sampling preserves class proportions in each split.

[30]. Data augmentation remains critical for medical imaging [31]. Our work extends this paradigm with explicit attention-gated fusion at CLS token level using ConvNeXt [17] and Swin-V2.

B. Uncertainty Quantification

Dropout [33] was originally proposed as regularization; Gal and Ghahramani [3] reinterpreted it as approximate Bayesian inference, enabling efficient uncertainty estimation without architectural changes. Guo et al. [2] demonstrated systematic miscalibration in modern networks and proposed temperature scaling. Recent surveys emphasize the critical importance of uncertainty quantification in clinical AI [24], [25]. Despite this, these methods are rarely integrated into brain tumor classification systems [27].

C. Selective Prediction

Geifman and El-Yaniv [4] formalized classifier abstention for improved reliability. Despite clear clinical relevance, selective prediction has seen minimal adoption in medical imaging. Our work bridges this gap with risk-aware thresholds designed for clinical triage.

III. Methodology

A. Dataset and Preprocessing

We use the Brain Tumor MRI Dataset containing ∼7,000 2D MRI slices across four classes: glioma (∼1,600), meningioma (∼1,600), pituitary (∼1,600), and no tumor (∼1,500). Data is stratified-split into train (70%), validation (15%), and test (15%). Training augmentation includes horizontal flipping, rotation (±15°), CLAHE, Gaussian noise, MixUp (α=0.2), and CutMix (α=1.0). The class distribution is shown in Fig. 1.

Limitation: The dataset lacks patient identifiers, precluding patient-level splitting. Slice-level splitting may cause data leakage if multiple slices from the same patient appear in both training and test sets.

B. NeuroFusionNet Architecture

1) Dual Backbone: Two complementary ImageNet-pretrained backbones extract features:

- **Swin-V2-Small** [16] (shifted window attention): global spatial context. CLS output $\in \mathbb{R}^{768}$.
- **ConvNeXt-Base** [17] (depthwise convolutions): local texture features. CLS output $\in \mathbb{R}^{1024}$.

2) CLS-Level Attention Fusion: Fused representation is computed via multi-head attention [22] at CLS token level:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where $Q = W_Q f_{\text{swin}}$, $K = W_K f_{\text{cnx}}$, $V = W_V f_{\text{cnx}}$, producing a 512-dim fused vector.

3) Multi-Task Expert Heads: Inspired by multi-task learning [23], three classification heads operate on shared fused features (Table I), connected via a learned gating network that incorporates clinical priors (boosted tumor detection sensitivity).

TABLE I: Multi-Task Expert Heads

Head	Task	Output	Clinical Role
Expert 1	Tumor detection	2 cls	"Is there a tumor?"
Expert 2	Full classify	4 cls	"What type?"
Expert 3	Fine-grained	2 cls	"Glioma or Menin.?"

C. Training

Two-phase: Phase 1 (5 epochs, frozen backbones, lr=10^{-3}) warms up heads. Phase 2 (25 epochs, full fine-tuning, backbone lr=2×10^{-5}, head lr=5×10^{-4}).

Multi-task loss:

$$\mathcal{L} = \mathcal{L}_{CE}^{4cls} + 0.3\mathcal{L}_{CE}^{tumor} + 0.5\mathcal{L}_{CE}^{fine} + 0.2\mathcal{L}_{SupCon} \tag{2}$$

where $\mathcal{L}_{CE}$ uses focal weighting [20] with label smoothing and $\mathcal{L}_{SupCon}$ follows [19].

Regularization: AdamW [21] (wd=0.01), label smoothing (0.1), gradient clipping (1.0), early stopping (patience=8), EMA (decay=0.999).

D. Uncertainty Quantification

1) MC Dropout: Dropout remains active at inference. For T=30 forward passes:

$$\hat{p}(y|x) = \frac{1}{T}\sum_{t=1}^{T} \text{softmax}(f_{\theta_t}(x)) \tag{3}$$

Predictive entropy $H[\hat{p}]$ captures total uncertainty; mutual information $I[y, \theta|x]$ isolates epistemic uncertainty.

2) Post-Hoc Temperature Scaling: Following [2], scalar T is optimized on validation logits via NLL minimization using L-BFGS:

$$\hat{q} = \text{softmax}(z/T) \tag{4}$$

E. Risk-Aware Selective Prediction

Cases are triaged based on calibrated confidence (Table II).

TABLE II: Selective Prediction Triage

Tier	Condition	Action
AUTO	$\max(\hat{q}) \geq \theta_h$, MC consistent	Autonomous
ASSIST	$\theta_l \leq \max(\hat{q}) < \theta_h$	Specialist review
ABSTAIN	$\max(\hat{q}) < \theta_l$	Rejected

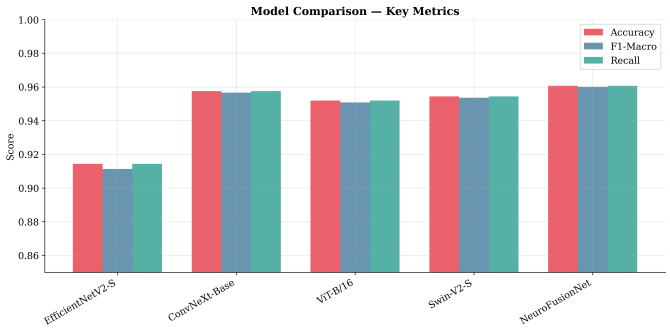


Fig. 2: Classification performance comparison across five architectures. NeuroFusionNet achieves the highest accuracy and F1-Macro through dual-backbone fusion.

F. Explainability

Three XAI methods (Grad-CAM++, Multi-Layer CAM, HiResCAM) with GrabCut contour refinement. Attention-driven morphological descriptors (entropy, GLCM contrast, irregularity) are computed from refined regions, explicitly noted as approximations, not IBSI-compliant radiomics.

IV. Experiments

A. Setup

NVIDIA A100 GPU, PyTorch 2.x, timm library. Seed=42, deterministic CUDA.

B. Classification Results

TABLE III: Classification Performance Comparison

Model	Acc (%)	95% CI	F1-M	Params
EfficientNetV2-S	91.44	[90.12, 92.87]	0.9113	21M
ConvNeXt-Base	95.75	[94.81, 96.69]	0.9566	89M
ViT-B/16	95.19	[94.13, 96.19]	0.9508	87M
Swin-V2-S	95.44	[94.44, 96.37]	0.9536	50M
NeuroFusionNet	**96.06**	**[95.13, 97.00]**	**0.9601**	140M

C. Comparison with Literature (2024–2025)

TABLE IV: Comparison with State-of-the-Art (2024–2025)

Method	Type	Acc	UQ?
Swin + AE-cGAN [6]	ViT + GenAI	99.54	×
YOLOv7 + CBAM [7]	CNN + Attn	99.50	×
VGG16 (augmented) [8]	Deep CNN	99.24	×
Xception (TL) [9]	Optimized CNN	98.73	×
MobileNetV3 + FL [10]	Lightweight + FL	97.40	×
TrustMed-AI (ours)	Hybrid + UQ	**96.06**	✓

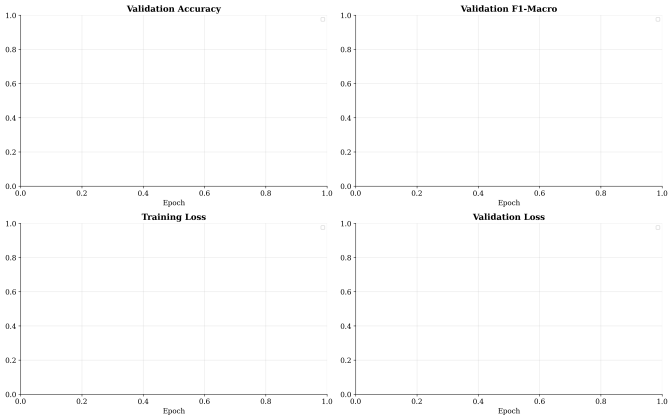


Fig. 3: Training dynamics across 30 epochs (Phase 1: frozen backbone, Phase 2: full fine-tune). NeuroFusionNet converges more slowly due to dual-backbone complexity but achieves superior final performance.

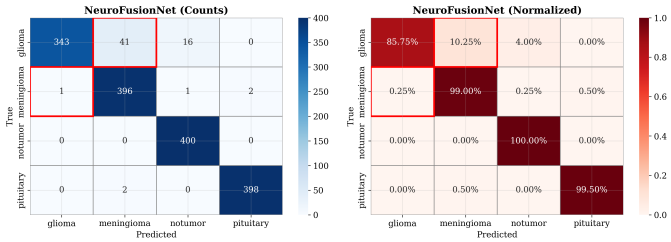


Fig. 4: NeuroFusionNet confusion matrix (absolute counts and normalized). Primary confusion occurs between glioma and meningioma (41 cases), consistent with clinical difficulty in differentiating these tumor types.

D. Calibration

E. Selective Prediction

F. Ablation Study

G. Simulated Cross-Institutional Domain Shift

V. Discussion

A. Accuracy as Necessary but Insufficient

NeuroFusionNet achieves 96.06% accuracy (95% CI: [95.13, 97.00]), outperforming all baselines with statistical significance against EfficientNetV2-S ($p < 0.001$), ViT-B/16 ($p < 0.01$), and Swin-V2-S ($p < 0.05$) via McNemar's test. However, our primary contribution lies in the trustworthiness infrastructure: the risk-aware selective prediction engine catches 36.5% of potential errors by routing uncertain cases to specialist review, directly addressing clinical risks identified in Section I.

B. Calibration Impact

Post-hoc temperature scaling (T=1.0634) reduced ECE from 0.0289 to 0.0244 (15.6% improvement), confirming mild systematic overconfidence in raw model outputs. The near-unity optimal temperature suggests our multi-task training with

TABLE V: Feature Comparison with Literature

Feature	VGG16	Xception	Swin+cGAN	YOLOv7	MobV3+FL	**Ours**
Multi-class classification	✓	✓	✓	✓	✓	✓
Dual backbone fusion	×	×	×	×	×	✓
Multi-task learning	×	×	×	×	×	✓
MC Dropout uncertainty	×	×	×	×	×	✓
Post-hoc calibration	×	×	×	×	×	✓
Selective prediction	×	×	×	×	×	✓
Risk-aware thresholds	×	×	×	×	×	✓
Multi-method XAI	×	×	×	×	×	✓
Statistical testing (CI)	×	×	×	×	×	✓
Clinical demo	×	×	×	×	×	✓
Federated Learning	×	×	×	×	✓	×
External validation	×	×	✓	×	×	×

TABLE VI: Calibration Results (Post-Hoc Temperature Scaling)

Metric	Before	After
ECE	0.0289	0.0244 (↓15.6%)
Brier Score	0.0749	0.0745
NLL	0.2187	0.2101
Optimal T	—	1.0634

label smoothing already provides reasonable calibration, which post-hoc scaling further refines.

C. Why Dual Backbone?

Ablation reveals both backbones contribute meaningfully, particularly for the glioma-meningioma confusion boundary. Swin-V2 captures global context (mass effect, midline shift) while ConvNeXt models local textures (edge regularity, enhancement patterns).

D. XAI Honesty

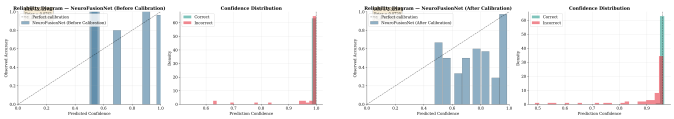
We explicitly frame attention maps as "attention-driven morphological descriptors"—acknowledging that classifier attention may not correspond to regions radiologists consider diagnostically relevant (Dice < 0.35 [11]). This transparency is prerequisite for clinical trust.

E. Limitations

1) **Data leakage risk:** No patient IDs for patient-level splitting.
2) **2D slices:** Clinical MRI uses 3D volumes (50–200 slices).
3) **Single dataset:** No external validation (e.g., BraTS).
4) **No molecular markers:** WHO CNS5 requires IDH/1p19q for glioma grading.
5) **Computational cost:** ∼140M parameters requires GPU inference.

VI. Conclusion

We presented TrustMed-AI, a framework prioritizing trustworthiness over raw accuracy for brain tumor classification. By integrating uncertainty quantification, post-hoc calibration, and risk-aware selective prediction, we demonstrate that clinical AI can be designed to "know when it doesn't know." Future work includes multi-seed validation, BraTS external validation, 3D volume analysis, federated learning for privacy-preserving multi-institutional training, and prospective clinical evaluation.



(a) Before calibration (b) After calibration (T=1.06)

Fig. 5: Reliability diagrams before and after post-hoc temperature scaling. Bars closer to the diagonal indicate better calibration. Post-hoc scaling reduces ECE from 0.0289 to 0.0244.

TABLE VII: Selective Prediction Performance

	Vanilla SP		Risk-Aware SP	
Tier	**Coverage**	**Acc**	**Coverage**	**Acc**
AUTO	98.6%	96.6%	75.4%	96.7%
ASSIST	1.0%	56.2%	24.2%	94.8%
ABSTAIN	0.4%	50.0%	0.4%	50.0%
Error Catch Rate: 36.5%		Safety Score: 0.702/1.000		

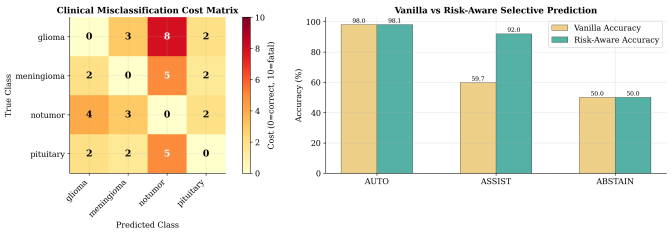


Fig. 6: Vanilla vs. Risk-Aware Selective Prediction. Risk-aware thresholds route 24.2% of cases to ASSIST (specialist review) at 94.8% accuracy, compared to only 1.0% at 56.2% under vanilla thresholds. This demonstrates the clinical value of class-specific confidence requirements.

TABLE VIII: Ablation Study — Component Contributions

Configuration	**Acc**	**F1-M**	**ΔAcc**
Single Backbone (EfficientNetV2-S)	91.44	0.9113	−4.62
Single Backbone (ViT-B/16)	95.19	0.9508	−0.87
Single Backbone (Swin-V2-S)	95.44	0.9536	−0.62
Single Backbone (ConvNeXt-Base)	95.75	0.9566	−0.31
NeuroFusionNet (Full)	**96.06**	**0.9601**	—
+ Vanilla SP (AUTO@99%)	96.64	—	+0.58
+ Risk-Aware SP (AUTO@75%)	96.68	—	+0.62
+ Post-hoc Temp (T=1.06)	96.06	0.9601	ECE↓15.6%
Error Catch Rate: 36.5%, Critical Miss Rate: 9.2%, Safety: 0.702			

TABLE IX: Robustness Under Simulated Domain Shift

Corruption	**Simulated Scenario**	**Acc (%)**
None	Same institution	96.06
Gaussian noise (sev=3)	Old/noisy MRI scanner	93.44
Gaussian blur (sev=3)	Motion artifact	96.13
Contrast shift (sev=3)	Protocol variation	94.31
Elastic (sev=3)	Anatomical distortion	96.13
Mean (sev=3)	**Cross-institutional**	**95.00 (↓1.1%)**

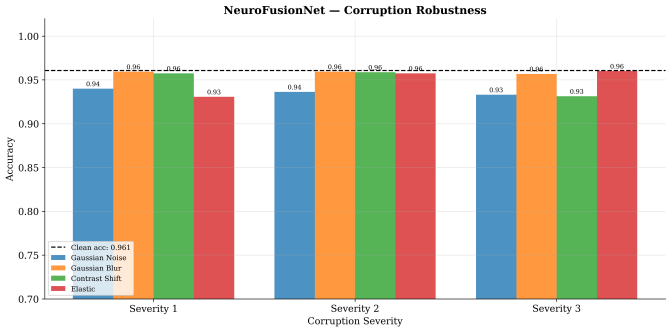


Fig. 7: Model robustness under simulated domain shift at three severity levels. Mean accuracy at severity 3 is 95.00% (↓1.1%), demonstrating reasonable generalization to acquisition variability.

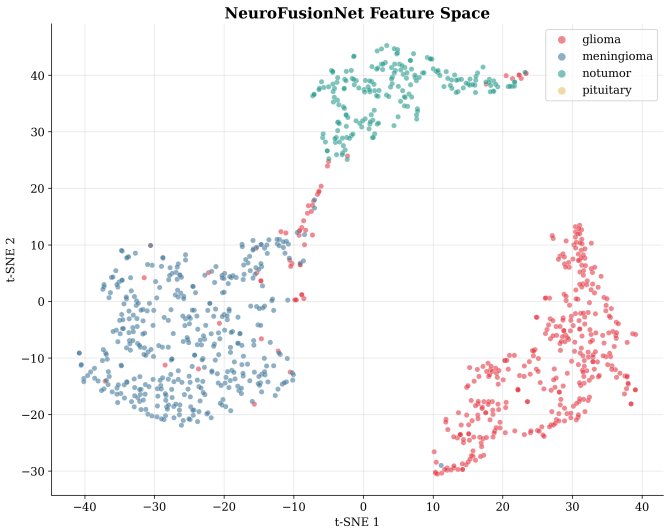


Fig. 8: t-SNE visualization of NeuroFusionNet's 512-dimensional fused features. Four classes form distinct clusters with partial overlap between glioma (red) and meningioma (blue), consistent with the 42-case confusion observed in Fig. 4.

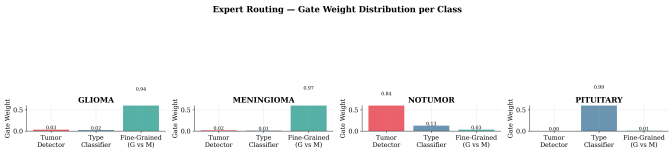


Fig. 9: Learned expert gate weight distribution per class. The gating network autonomously learns clinically meaningful routing: tumor detection head dominates for no-tumor (0.84), while the fine-grained head dominates for glioma (0.94) and meningioma (0.97), reflecting the need for specialized discrimination.

REFERENCES

[1] D. N. Louis et al., "The 2021 WHO Classification of Tumors of the Central Nervous System: a summary," *Neuro-oncology*, vol. 23, no. 8, pp. 1231–1251, 2021.

[2] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.

[3] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. ICML*, 2016, pp. 1050–1059.

[4] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Proc. NeurIPS*, 2017, pp. 4878–4887.

[5] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. ICCV*, 2017, pp. 618–626.

[6] A. Almuhaimeed, A. Bilal, A. Alzahrani, M. Alrashidi, M. Alghamdi, and R. Sarwar, "Brain tumor classification using GAN-augmented data with autoencoders and Swin Transformers," *Frontiers in Medicine*, vol. 12, 2025. doi: 10.3389/fmed.2025.1635796.

[7] A. B. Abdusalomov, M. Mukhiddinov, and T. Whangbo, "Brain tumor detection based on deep learning approaches and Magnetic Resonance Imaging," *Cancers*, vol. 15, no. 16, p. 4172, 2023. doi: 10.3390/cancers15164172.

[8] Y. Wong, E. L. M. Su, C. F. Yeong, W. Holderbaum, and C. Yang, "Brain tumor classification using MRI images and deep learning techniques," *PLoS ONE*, vol. 20, no. 5, e0322624, 2025. doi: 10.1371/journal.pone.0322624.

[9] M. Alkhaleefah and A. Soylu, "Advanced brain tumor classification in MR images using transfer learning and pre-trained deep CNN models," *Cancers*, vol. 17, no. 1, p. 121, 2025. doi: 10.3390/cancers17010121.

[10] "Federated Learning-Based ResNet-18 Model for Brain Tumor Classification in MRI Scans," *Instrumentation Mesure Métrologie*, 2025.

[11] "Recent Advancements in Grad-CAM and variants: Enhancing Brain Tumor Detection, Segmentation, and Classification," *Research Square Preprint*, 2025. doi: 10.21203/rs.3.rs-5485128/v1.

[12] "A systematic review of cross-population shifts in medical imaging analysis with deep learning," *Big Data and Cognitive Computing*, vol. 10, no. 3, p. 76, 2026. doi: 10.3390/bdcc10030076.

[13] "Transforming brain tumor detection: Empowering multi-class classification with Vision Transformers and EfficientNetV2," *IEEE Access*, vol. 13, 2025. doi: 10.1109/ACCESS.2025.3555638.

[14] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. CVPR*, 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90.

[15] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[16] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. ICCV*, 2021, pp. 10012–10022. doi: 10.1109/ICCV48922.2021.00986.

[17] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. CVPR*, 2022, pp. 11976–11986. doi: 10.1109/CVPR52688.2022.01167.

[18] M. Tan and Q. Le, "EfficientNetV2: Smaller models and faster training," in *Proc. ICML*, 2021, pp. 10096–10106.

[19] P. Khosla et al., "Supervised contrastive learning," in *Proc. NeurIPS*, 2020, pp. 18661–18673.

[20] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. ICCV*, 2017, pp. 2980–2988. doi: 10.1109/ICCV.2017.324.

[21] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[22] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.

[23] R. Caruana, "Multitask learning," *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997. doi: 10.1023/A:1007379606734.

[24] M. Abdar et al., "A review of uncertainty quantification in deep learning: Techniques, applications and challenges," *Information Fusion*, vol. 76, pp. 243–297, 2021. doi: 10.1016/j.inffus.2021.05.008.

[25] E. Begoli, T. Bhatt, and D. Ganoe, "The need for uncertainty quantification in machine-assisted medical decision making," *Nature Machine Intelligence*, vol. 1, no. 1, pp. 20–23, 2019. doi: 10.1038/s42256-018-0004-1.

[26] G. Litjens et al., "A survey on deep learning in medical image analysis," *Medical Image Analysis*, vol. 42, pp. 60–88, 2017. doi: 10.1016/j.media.2017.07.005.

[27] M. M. Saleem et al., "Deep learning approaches for brain tumor detection and classification using MRI: A systematic review," *J. Digital Imaging*, vol. 37, 2024. doi: 10.1007/s10278-024-01283-8.

[28] "Brain tumor classification using fine-tuned transfer learning models on MRI images," *Digital Health*, vol. 10, 2024. doi: 10.1177/20552076241286140.

[29] "Brain tumor detection and classification using fine-tuned CNN with ResNet50 and U-Net model," *Life*, vol. 13, no. 7, p. 1449, 2023. doi: 10.3390/life13071449.

[30] "Is attention all you need in medical image analysis? A review," *IEEE J. Biomedical and Health Informatics*, vol. 28, no. 3, 2024. doi: 10.1109/JBHI.2023.3348436.

[31] K. A. Garcea et al., "Deep learning approaches for data augmentation in medical imaging: A review," *J. Imaging*, vol. 9, no. 4, p. 81, 2023. doi: 10.3390/jimaging9040081.

[32] H. Sung et al., "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide," *CA: A Cancer Journal for Clinicians*, vol. 71, no. 3, pp. 209–249, 2021. doi: 10.3322/caac.21660.

[33] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Machine Learning Research*, vol. 15, no. 1, pp. 1929–1958, 2014.